

Discrete-time state feedback control with full-order observer

1 Introduction

1. Scelta controllore:

- (a) Tracking asintotico
 - i. PID o PID+Antiwindup
 - ii. State-Space approach $\implies$ Nominal tracking
- (b) Tracking robusto $\implies$ State-Space approach $\implies$ Robust tracking
 - i. Robust tracking with integral action
 - ii. Robust tracking with internal model principle
 - A. Robust tracking with error-space approach
 - B. Robust tracking with extended estimator approach
- (c) Tracking ottimo $\implies$ LQR

2. Definizione specifiche Tr, Mp

$$\begin{aligned} \text{Tr} &= 0.25; \\ \text{Mp} &= 0.2; \end{aligned}$$

3. Calcolo sample time

$$T_s \leq \frac{1}{10} \cdot \frac{2\pi}{\omega_B} = \frac{2\pi}{10 \cdot \omega_n}$$

$$\omega_n = 1.8/\text{Tr};$$

$$T_{s_max} = (2\pi)/(10\omega_n);$$

$$T_s = T_{s_max} / 4;$$

4. Definizione P(s)

- (a) Date le matrici A,B,C,D

$$\text{ssP} = \text{ss}(A,B,C,D);$$

$$P = \text{tf}(\text{ssP});$$

- (b) dato P(s)

$$s = \text{tf}('s');$$

$$P = 90/(s^2 + 60s + 3);$$

5. Calcolo P(z)

$$P_d = \text{c2d}(P,T_s,'zoh');$$

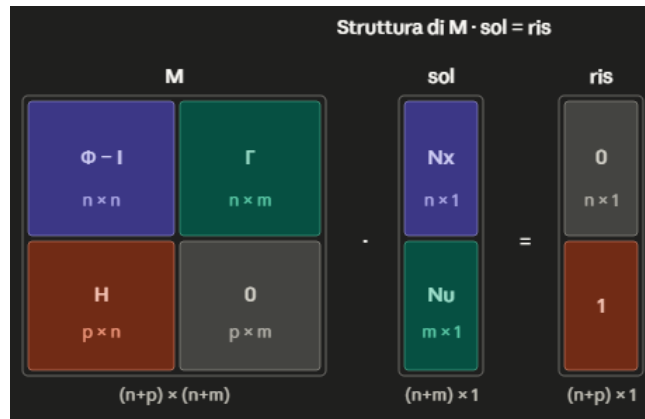
6. State-space discrete model

```
ssP_d = ss(P_d);
```

7. Calcolo matrici discrete

```
[Phi,Gamma,H,J] = ssdata(ssP_d);
```

8. Calcolo di Nx e Nu



```
n = size(Phi, 1);    % ordine del sistema
p = size(H, 1);      % numero di uscite
m = size(Gamma, 2);  % numero di ingressi
```

```
M = [Phi-eye(n), Gamma; H, zeros(p,m)];
```

```
ris = [zeros(n, 1); 1];
```

```
sol = inv(M) * ris;
```

```
Nx = sol(1:n);
Nu = sol(n+1:end);
```

9. Piazzamento dei poli

$$p_{c1,2} = -\delta\omega_n \pm j\omega_n\sqrt{1-\delta^2}$$

$$\delta = \frac{\log(1/M_p)}{\sqrt{\pi^2 + \log^2(1/M_p)}}$$

$$p_{d1,2} = e^{p_{c1,2} * T_s}$$

```
delta = log(1/Mp) / (sqrt(pi^2 + log(1/Mp)^2));
```

```
p_c_1 = -omega_n*delta + 1i * omega_n * sqrt(1-delta^2); %poli nel continuo
p_c_2 = conj(p_c_1);
```

```
p_d = [exp(p_c_1*Ts),exp(p_c_2*Ts)]; %poli nel discreto
```

10. Calcolo di K

```
K = place(Phi,Gamma,p_d);
```

11. Full-state observer

```
poles_obs_z = [exp(-40*Ts),exp(-50*Ts)];
```

%Piazzamento poli observer molto prima dei
% poli della catena chiusa

```
L = place(Phi',H',poles_obs_z)';
```

%Gain observer

```
Phi_obs = Phi-L*H;
```

%Matrici observer

```
Gamma_obs = [Gamma, L];
```

```
H_obs = eye(n);
```

```
J_obs = zeros(n, m+p);
```

12. Simulink model

